

March 4, 2026

# Some exercises on differentiable paths

Proposition: If  $\vec{x}, \vec{y} : I \rightarrow \mathbb{R}^n$  are differentiable paths,  
then 
$$\frac{d}{dt}(\vec{x} \cdot \vec{y}) = \frac{d\vec{x}}{dt} \cdot \vec{y} + \vec{x} \cdot \frac{d\vec{y}}{dt}$$

proof: Let  $\vec{x}(t) = (x_1(t), \dots, x_n(t))$  and  $\vec{y}(t) = (y_1(t), \dots, y_n(t))$ .

Let  $x_1, \dots, x_n, y_1, \dots, y_n : I \rightarrow \mathbb{R}$  are differentiable functions.

Now, 
$$\vec{x} \cdot \vec{y}(t) = \vec{x}(t) \cdot \vec{y}(t) = \sum_{i=1}^n x_i(t) y_i(t) \quad \text{and}$$

$$\begin{aligned} \frac{d}{dt}(\vec{x} \cdot \vec{y}) &= \frac{d}{dt} \left( \sum_{i=1}^n x_i(t) y_i(t) \right) = \sum_{i=1}^n \frac{d}{dt} (x_i(t) y_i(t)) \\ &= \sum_{i=1}^n \left( \frac{dx_i}{dt}(t) y_i(t) + x_i(t) \frac{dy_i}{dt}(t) \right) \\ &= \left( \sum_{i=1}^n \frac{dx_i}{dt}(t) y_i(t) \right) + \left( \sum_{i=1}^n x_i(t) \frac{dy_i}{dt}(t) \right) = \frac{d\vec{x}}{dt}(t) \cdot \vec{y}(t) + \vec{x}(t) \cdot \frac{d\vec{y}}{dt}(t) \end{aligned}$$

Proposition: If  $\vec{u}, \vec{v} : I \rightarrow \mathbb{R}^3$  are differentiable paths,  
then 
$$\frac{d}{dt}(\vec{u} \times \vec{v}) = \frac{d\vec{u}}{dt} \times \vec{v} + \vec{u} \times \frac{d\vec{v}}{dt}$$

proof: Similar to the one above, but longer.



Proposition: If  $\vec{x} : I \rightarrow \mathbb{R}^n$  is a differentiable path and  $\vec{x}$  satisfies  $\|\vec{x}(t)\|$  is constant for all  $t$ , then  $\vec{x} \cdot \vec{v} = 0$  for all  $t$ .

proof:  $\|\vec{x}(t)\| = \sqrt{\sum_{i=1}^n (x_i(t))^2} = \sqrt{\vec{x} \cdot \vec{x}}$  
$$\frac{d}{dt}(\vec{x} \cdot \vec{x}) = 0 = \frac{d\vec{x}}{dt} \cdot \vec{x} + \vec{x} \cdot \frac{d\vec{x}}{dt} = 2 \vec{x} \cdot \vec{v}.$$

Corollary: Assume  $\vec{x} : I \rightarrow \mathbb{R}^n$  is a twice differentiable path.

If at  $t = t_0$  we have  $\frac{d}{dt}(\|\vec{v}\|)(t_0) = 0$ , then  $(\vec{v} \cdot \vec{a})(t_0) = 0$ .

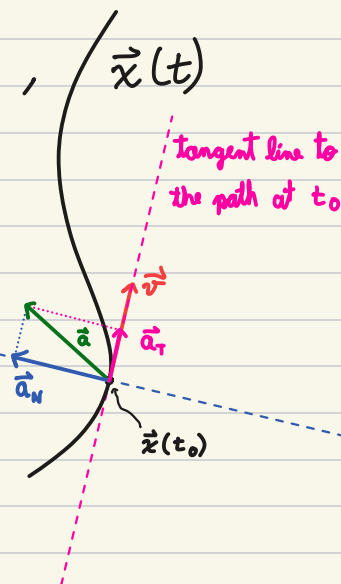
proof: Same as above, i.e.  $\frac{d}{dt}\|\vec{v}\| = 0 \Rightarrow \frac{d}{dt}(\vec{v} \cdot \vec{v}) = 0 = \frac{d\vec{v}}{dt} \cdot \vec{v} + \vec{v} \cdot \frac{d\vec{v}}{dt} = 2 \vec{v} \cdot \vec{a}$

## Tangential and normal components of acceleration:

When the rate of change of speed ( $\frac{d}{dt} \|\vec{v}\|$ ) is 0, the acceleration  $\vec{a}$  is perpendicular to the velocity  $\vec{v}$ .

Conversely, if the acceleration is perpendicular to the velocity, then the rate of change of speed is 0.

line perpendicular  
(or normal) to  
the tangent line



def: The tangential component of the acceleration is defined as  $\vec{a}_T = \text{proj}_{\vec{v}} \vec{a}$  whenever  $\vec{v}(t) \neq 0$ .

(remark:  $\vec{v}$  and  $\vec{a}$  depend on time  $t$ , so  $\vec{a}_T$  also depends on time).

The normal component of the acceleration is given by

$$\vec{a}_N = \vec{a} - \vec{a}_T = \vec{a} - \text{proj}_{\vec{v}} \vec{a}.$$

$$\begin{aligned} * \text{proj}_{\vec{v}} \vec{a} &= \frac{\vec{a} \cdot \vec{v}}{\vec{v} \cdot \vec{v}} \vec{v} \\ &= (\vec{a} \cdot \hat{v}) \hat{v} \end{aligned}$$

proposition: The rate of change of the speed ( $\frac{d}{dt} \|\vec{v}\|$ ) satisfies  $\vec{a}_T = \left( \frac{d}{dt} \|\vec{v}\| \right) \frac{1}{\|\vec{v}\|} \vec{v}$  whenever  $\vec{v} \neq 0$  and  $\frac{d\|\vec{v}\|}{dt} = \frac{\vec{a} \cdot \vec{v}}{\|\vec{v}\|}$ .

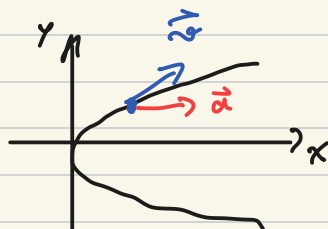
proof: By definition,  $\|\vec{v}\| = \sqrt{\vec{v} \cdot \vec{v}}$ , so  $\frac{d}{dt} (\|\vec{v}\|^2) = 2 \|\vec{v}\| \frac{d\|\vec{v}\|}{dt}$ .  
also  $\frac{d}{dt} (\|\vec{v}\|^2) = \frac{d}{dt} (\vec{v} \cdot \vec{v}) = \frac{d\vec{v}}{dt} \cdot \vec{v} + \vec{v} \cdot \frac{d\vec{v}}{dt} = 2 \frac{d\vec{v}}{dt} \cdot \vec{v} = 2 \vec{a} \cdot \vec{v}$ .  
Thus,  $\frac{d}{dt} \|\vec{v}\| = \frac{\vec{a} \cdot \vec{v}}{\|\vec{v}\|}$ . On the other hand  
 $\vec{a}_T = \frac{\vec{v} \cdot \vec{a}}{\vec{v} \cdot \vec{v}} \vec{v} = \frac{\vec{a} \cdot \vec{v}}{\|\vec{v}\|^2} \vec{v} = \left( \frac{d}{dt} \|\vec{v}\| \right) \frac{1}{\|\vec{v}\|} \vec{v}$

e.g.: Find the tangential and normal components for the path  
also find the rate of change of speed at  $t = t_0$

$$\vec{x}(t) = t^2 \hat{i} + t \hat{j} \quad t_0 = 2$$

$$\vec{v} = \frac{d\vec{x}}{dt} = 2t \hat{i} + \hat{j}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = 2 \hat{i}$$



$$\text{tangential comp: } \vec{a}_T = \frac{\vec{a} \cdot \vec{v}}{\vec{v} \cdot \vec{v}} \vec{v} = \frac{4t}{(2t)^2 + 1} (2t \hat{i} + \hat{j}) = \frac{4t}{4t^2 + 1} (2t \hat{i} + \hat{j})$$

$$= \left( \frac{8t^2}{4t^2 + 1}, \frac{4t}{4t^2 + 1} \right)$$

$$\text{normal comp: } \vec{a}_N = \vec{a} - \vec{a}_T = 2 \hat{i} - \left( \frac{8t^2}{4t^2 + 1} \hat{i} + \frac{4t}{4t^2 + 1} \hat{j} \right)$$

$$= \left( \frac{2(4t^2 + 1)}{4t^2 + 1} - \frac{8t^2}{4t^2 + 1} \right) \hat{i} - \frac{4t}{4t^2 + 1} \hat{j}$$

$$= \left( \frac{2}{4t^2 + 1}, \frac{-4t}{4t^2 + 1} \right)$$

$$\text{rate of change of speed at } t=2: \frac{d\|\vec{v}\|}{dt} = \frac{\vec{a} \cdot \vec{v}}{\|\vec{v}\|} = \frac{4t}{\sqrt{4t^2 + 1}} = \frac{8}{\sqrt{16 + 1}}$$

$$\vec{x}(t) = (4 \cos t, 4 \sin t, 3t) \quad t_0 = 3/4$$

$$\vec{v} = \frac{d\vec{x}}{dt} = (-4 \sin t, 4 \cos t, 3) \quad \vec{a} = \frac{d\vec{v}}{dt} = (-4 \cos t, -4 \sin t, 0)$$

$$\text{tangential: } \vec{a}_T = \frac{\vec{a} \cdot \vec{v}}{\vec{v} \cdot \vec{v}} \vec{v} = \frac{16 \sin t \cos t - 16 \cos t \sin t}{\vec{v} \cdot \vec{v}} \vec{v} = 0$$

$$\vec{a}_N = \vec{a}$$

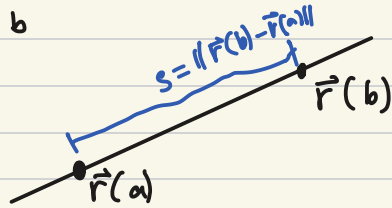
$$\text{rate of change of speed for any } t: \frac{d\|\vec{v}\|}{dt} = \frac{\vec{a} \cdot \vec{v}}{\|\vec{v}\|} = 0$$

# Arclength

Distance traveled along a rectilinear path at constant velocity:

$$\vec{r}(t) = \vec{r}_0 + t \vec{v} \quad \text{for } a \leq t \leq b$$

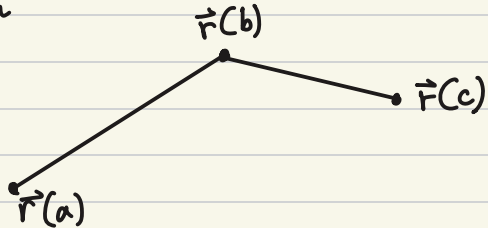
$$\begin{aligned} \text{distance} &= s = \|\vec{r}(b) - \vec{r}(a)\| \\ &= \|b \vec{v} - a \vec{v}\| \\ &= (b-a) \|\vec{v}\| \end{aligned}$$



Now piece together two linear paths

$$a \leq t \leq b \quad a \leq b \leq c$$

$$\vec{r}(t) = \begin{cases} \vec{r}_0 + t \vec{v}_1, & a \leq t \leq b \\ \vec{r}_b + t \vec{v}_2, & b \leq t \leq c \end{cases}$$



$$\begin{aligned} s &= \|\vec{r}(b) - \vec{r}(a)\| + \|\vec{r}(c) - \vec{r}(b)\| \\ &= (b-a) \|\vec{v}_1\| + (c-b) \|\vec{v}_2\| \end{aligned}$$

$$\vec{v}_1 = \frac{d\vec{r}}{dt}(t_1) \quad a < t_1 < b$$

$$\vec{v}_2 = \frac{d\vec{r}}{dt}(t_2) \quad b < t_2 < c$$

$$\begin{aligned} s &= (b-a) \|\vec{v}_1\| + (c-b) \|\vec{v}_2\| = \int_a^b \|\vec{v}_1\| dt + \int_b^c \|\vec{v}_2\| dt \\ &= \int_a^c \|\vec{v}\| dt \end{aligned}$$

General path

differentiable path  $\vec{x}: [a, b] \rightarrow \mathbb{R}^n$

$$\text{total length } s = \int_a^b \|\vec{v}\| dt$$

$$\begin{aligned} \text{e.g. } \vec{x}(t) &= (\cos t, \sin t) \quad 0 \leq t \leq \theta \\ \vec{v}(t) &= (-\sin t, \cos t) \quad \|\vec{v}\| = 1 \end{aligned}$$

$$\begin{aligned} \int_0^\theta \|\vec{v}\| dt &= \int_0^\theta 1 dt \\ &= t \Big|_0^\theta = \theta \end{aligned}$$

